Notes: Arieli, Babichenko, Peretz, and Young (Econometrica, 2020)

The Speed of Innovation Diffusion in Social Networks

Contents

1 Big picture

- **Question.** How fast can a superior innovation spread through a social network when the network structure is unknown, directed, irregular, or changing over time?
- **Main contribution.** The paper proves a universal upper bound on expected waiting time until a target fraction of agents adopts. The bound does not depend on the network topology, network size, degree distribution, or the speed at which links change.
- **Economic mechanism.** Adoption is governed by perturbed best responses in a coordination game. Even when nobody has adopted, random payoff shocks generate a positive probability that some agents switch to the innovation. These early switches can then spread by imitation and coordination.
- **Key technical object.** The response function $r(x)$ gives the probability that an agent adopts the innovation when a weighted fraction x of her neighbors already adopted.
- **Key condition.** The supporting line $L(x)$ below $r(x)$ must have positive intercept $L(0) > 0$. This means the innovation has enough initial "spontaneous adoption" pressure to avoid being trapped near zero adoption.
- **Main theorem.** If $L(0) > 0$ and the target q is below the fixed point p of the supporting line, the expected waiting time satisfies a topology-free bound proportional to

$$\frac{1}{L(0)}\left(1 + \ln\frac{p}{p-q}\right).$$

- **Why this is strong.** Existing bounds typically require regular, undirected, fixed networks. This result works for arbitrary directed weighted networks and time-varying networks.
- **Multiple innovations.** The same logic extends to many competing technologies by replacing $r(x)$ with a worst-case adoption probability $\rho(x)$ for the best technology.
- **Topology still matters.** More information about the network can yield sharper results. Large regular networks converge at exactly $\alpha^{-1}\ln(p/(p-q))$ under linear dynamics. Star networks have fast arrival but weak persistence.
- **Economic takeaway.** Diffusion speed can be bounded even without knowing the network. What matters for a universal bound is the response function's lower linear support, not the detailed topology.

2 Objects and measurement in the model

2.1 Network

A weighted directed network with m agents is represented by an $m \times m$ row-stochastic matrix

$$P(t) = \{P_{ij}(t)\}.$$

Here $P_{ij}(t)$ is the probability, or interaction weight, that agent i places on agent j at time t.

- The network can be directed or undirected.
- It can have arbitrary size and arbitrary degree distribution.
- It can vary over time.
- Self-weight $P_{ii}(t) > 0$ is allowed and can be interpreted as inertia.

Interpretation: The model is intentionally agnostic about network topology. This is why the main result is useful when social links are hard to observe.

2.2 Actions and payoff matrix

Initially there are two actions: status quo 0 and innovation 1. Payoffs from interacting with one neighbor are

	0	1
0	c	0
1	a	$a+c$

where

- $a > 0$ is the inherent payoff advantage of the innovation;
- $c > 0$ is the coordination payoff from matching the other agent's action.

Interpretation: If $c < a$, the innovation is a dominant strategy. If $c > a$, both all-status-quo and all-innovation can be pure equilibria. The interesting case is when the innovation is better but coordination makes the inferior status quo sticky.

2.3 State and neighbor adoption share

The state is

$$s(t) = (s_1(t), \dots, s_m(t)) \in \{0,1\}^m,$$

where $s_i(t) = 1$ means that agent i adopts the innovation. Starting point:

$$s(0) = (0, \dots, 0).$$

When agent i revises, the weighted share of her neighbors who adopted is

$$x_i(t) = \sum_{j \in [m]} P_{ij}(t) s_j(t).$$

Interpretation: This is the local social signal faced by agent i. It can differ across agents and over time.

2.4 Perturbed best response and response function

Each agent receives revision opportunities according to independent Poisson processes with rate one. At a revision time, the payoff difference between action 1 and action 0 is perturbed by an i.i.d. shock $\varepsilon_i(t)$ with c.d.f. F.

Conditional on $x_i(t)$, the probability of adopting is

$$\Pr[s_i(t) = 1 \mid x_i(t)] = \Pr[\varepsilon_i(t) > c - a - 2cx_i(t)] = 1 - F(c - a - 2cx_i(t)).$$

Define

$$r(x) = 1 - F(c - a - 2cx).$$

Interpretation: The response function summarizes the whole behavioral rule. It is increasing: more adopting neighbors make adoption more likely.

2.5 Supporting line

The supporting line $L(x)$ is the line below $r(x)$ that maximizes its intercept while satisfying

$$L(1) = r(1), \qquad L(x) \le r(x) \quad \text{for all } x \in [0,1].$$

Write

$$L(x) = p\alpha + (1-\alpha)x,$$

where $1-\alpha$ is the slope and $p = L(0)/\alpha$ is the fixed point of L.

Equivalently,

$$L(0) = \inf_{x \in [0,1)} \frac{r(x) - xr(1)}{1-x}.$$

Interpretation: The supporting line is a lower bound on the actual adoption probability. If the lower-bound process diffuses quickly, the actual process diffuses at least as fast.

2.6 Target and waiting time

For a target adoption share $q \in [0,1]$, the expected waiting time is

$$T_q(F, P(t)) = \mathbb{E}\left[\min\left\{t : \frac{1}{m}\sum_{i\in[m]} s_i(t) \geq q,\ s(0) = (0,\ldots,0)\right\}\right].$$

Interpretation: The paper bounds the time needed to reach a population-level adoption target, not necessarily complete adoption.

3 Models and theoretical results

3.1 Main theorem: a network-free waiting-time bound

If $L(0) > 0$, L has slope $1-\alpha$, fixed point p, and $q < p$, then for every dynamic network $P(t)$,

$$T_q(F, P(t)) \leq \frac{7.2}{p\alpha}\left(1 + \ln\frac{p}{p-q}\right) = \frac{7.2}{L(0)}\left(1 + \ln\frac{p}{p-q}\right).$$

Interpretation: The right-hand side contains no network term. It depends only on the response function through $L(0)$ and p, and on the target q.

3.2 Meaning of the condition $L(0) > 0$

The condition $L(0) > 0$ holds when the response function is not too bowed near zero. For response functions generated by payoff shocks:

$$r(x) = 1 - F(c - a - 2cx).$$

If a rises, the response function shifts in favor of adoption. If the shock variance rises, spontaneous experimentation becomes more likely. Both forces can make $L(0) > 0$.

Interpretation: Economically, $L(0) > 0$ requires a nontrivial chance of adoption even before many neighbors adopt. Without this initial pressure, diffusion can be trapped near the status quo.

3.3 Proof intuition: lower-bound imitation process

Because $L(x) \leq r(x)$, it is enough to study the slower linear process

$$L(x) = p\alpha + (1-\alpha)x.$$

This process has a simple interpretation:

- with probability α, an updating agent tosses a coin and adopts with probability p;
- with probability $1-\alpha$, the agent imitates a neighbor drawn according to $P_i(t)$.

Interpretation: The proof converts a general perturbed best-response dynamic into an imitation-plus-experimentation dynamic. Experimentation plants adoption seeds; imitation spreads them through whatever network exists.

3.4 Imitation forest and Feige's inequality

A history of the imitation process defines an imitation forest. Roots are either original time-zero states or later coin-toss events. Agents at time t inherit actions through chains of imitation.

The key probability inequality is: if $c_1, \ldots, c_k$ are i.i.d. Bernoulli(p) and weights $\beta_i \geq 0$, then

$$\Pr\left(\sum_{i=1}^{k} \beta_i c_i \geq \left(\sum_i \beta_i\right) p\right) \geq 0.14p.$$

Interpretation: Even with arbitrary weights induced by a complicated changing network, there is a nonzero lower bound on the probability that the weighted average of coin-toss roots reaches its expectation.

3.5 Multiple competing technologies

Suppose actions are $A = \{1, \ldots, k\}$ and technology 1 is best, with $a_1 > a_2 > \cdots > a_k$. Payoffs take the form

$$U = \begin{array}{c|cccc} & 1 & 2 & \cdots & k \\ \hline 1 & a_1 + c & a_1 & \cdots & a_1 \\ 2 & a_2 & a_2 + c & \cdots & a_2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ k & a_k & a_k & \cdots & a_k + c \end{array}.$$

Let $r_i(\mathbf{x})$ be the probability technology i is the perturbed best response. For technology 1 define the worst-case adoption probability

$$\rho(x) = \min_{\mathbf{x} \in \Delta^k :\, x_1 = x} r_1(\mathbf{x}).$$

The two-action theorem applies to $\rho(x)$. If the shock distribution is log concave,

$$\rho(x) = r_1(x, 1 - x, 0, \ldots, 0).$$

Interpretation: The hardest competitor for the best technology is the second-best technology when shocks are log-concave. This reduces the multi-technology problem to a two-technology lower-bound problem.

3.6 Network-specific convergence rates

For the linear dynamic on large regular networks,

$$\lim_{m \to \infty} T_q(R_m) = \frac{1}{\alpha} \ln \frac{p}{p - q}.$$

This is tighter than the universal bound because regular networks have enough symmetry to permit a deterministic approximation.

For star networks, if $q < r(1)$,

$$\limsup_{m \to \infty} T_q(S_m) \leq \frac{r(1)}{r(1) - q} \left[\frac{1}{r(0)} + \ln\left(\frac{r(1)}{r(1) - q}\right)\right].$$

Interpretation: A star can diffuse quickly because once the hub adopts, many spokes respond as though their local adoption share is one.

3.7 Persistence and slow convergence

Let $I_{q,q'}(G)$ be the minimal expected time to fall to adoption share q' or below, starting from any state with adoption share at least q, where $q' < q$.

- For regular networks and $0 < q' < q < p$, persistence is strong:

$$I_{q,q'}(R_m) \geq c^m$$

  for some $c > 1$.

- For star networks with $r(1) < 1$ and $r(0) < 1/2$, persistence fails:

$$I_{1,1/2}(S_m) \leq c$$

  for a constant independent of m.

- If $r(q) < q$ in regular networks with degree d_m, convergence can be exponentially slow:

$$T_q(R_m) \geq \frac{e^{cd_m}}{m}.$$

Interpretation: Topology matters for sharper conclusions. Regular networks are persistent once adoption is high, while star networks can spread fast but can also lose adoption quickly if the hub switches back.

4 Identification and proof design

4.1 What is being identified?

This is a theory paper, not an empirical identification paper. The object is a worst-case bound on diffusion time that holds for every possible network process $P(t)$.

Interpretation: The result identifies sufficient behavioral conditions under which fast diffusion is guaranteed even without observing network topology.

4.2 Why the supporting line is central

- The actual response function $r(x)$ can be nonlinear and difficult to analyze directly.
- The supporting line $L(x)$ is linear and below $r(x)$ everywhere.
- A diffusion process based on L is slower than the actual process.
- Therefore a waiting-time bound for the L-process is also a bound for the original process.

Interpretation: This lower-bound argument turns a nonlinear coordination dynamic into a tractable imitation model.

4.3 Why the network disappears from the main bound

- The imitation forest tracks where agents' actions come from, even when links change over time.
- The weights on past coin tosses may be complicated, but they are nonnegative and sum to the mass of coin-toss descendants.
- Feige's inequality gives a uniform probability bound for weighted sums of independent Bernoulli variables.
- This uniform inequality is independent of network topology.

Interpretation: The proof avoids describing the whole network path. It only needs the existence of enough coin-toss descendants and a general probability bound for weighted Bernoulli sums.

4.4 What the paper cannot tell us

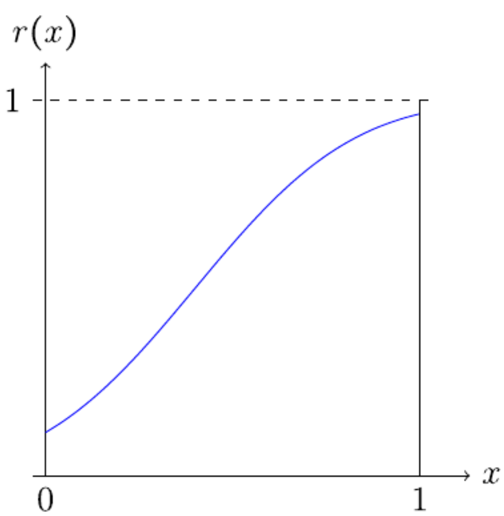
- The bound is sufficient, not necessary.
- The universal bound may be loose for known network classes.
- It does not estimate response functions from data.
- It assumes payoff shocks are i.i.d. over agents and time.
- It focuses on payoff-known coordination dynamics, not learning about unknown payoffs.
- It gives adoption-time guarantees, not welfare-maximizing policy rules for seeding or subsidies.

Interpretation: The result is powerful because it is topology-free, but that generality comes at the cost of tightness and empirical specificity.

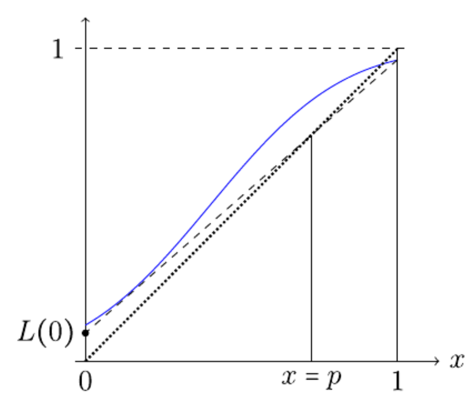
5 Figures and visual notes

The paper has no empirical tables. Its key visual material consists of six theoretical figures. They are grouped below to save space and improve comparison.

5.1 Figures 1 and 2: response function and supporting line



(a) Figure 1: response function under normal shocks.



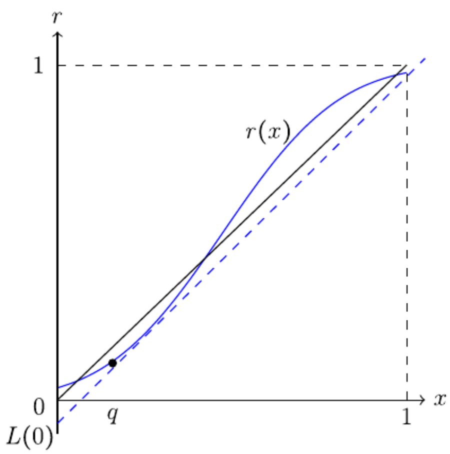
(b) Figure 2: supporting line $L(x)$ below $r(x)$.

Read:

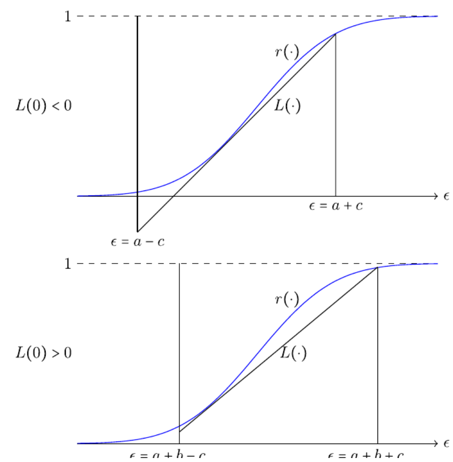
- Figure 1 shows an increasing response function: adoption probability rises with the local share of adopters.
- Figure 2 constructs the supporting line $L(x)$ below $r(x)$ with positive intercept $L(0)$.
- The fixed point p of L is below the fixed point of the full response function.

Economic message: The supporting line is the key lower-bound device. If the slower linear process reaches q, the actual nonlinear process can only be faster.

5.2 Figures 3 and 4: why $L(0) > 0$ matters



(a) Figure 3: low-level trap when $L(0) < 0$.



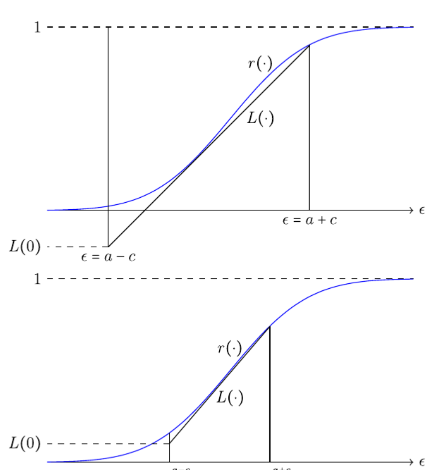
(b) Figure 4: increasing innovation payoff shifts $L(0)$ upward.

Read:

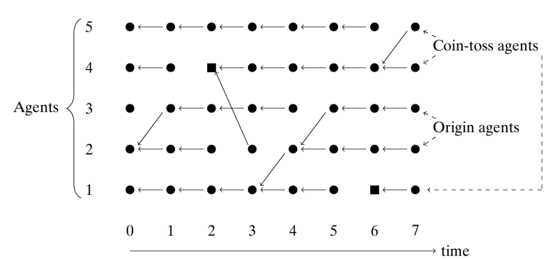
- Figure 3 shows a response function that is too bowed near zero. A low adoption share q can satisfy $r(q) < q$, creating a bottleneck.
- Figure 4 shows that raising the payoff advantage of the innovation can make the supporting-line intercept positive.

Economic message: Innovations diffuse quickly only when there is enough initial pressure to escape the status quo basin. Larger payoff advantages help create that pressure.

5.3 Figures 5 and 6: shock variance and imitation forest



(a) Figure 5: more dispersed shocks can increase $L(0)$.



(b) Figure 6: imitation forest representation of the proof.

Read:

- Figure 5 shows that larger shock dispersion can raise the intercept $L(0)$ by making spontaneous adoption more likely.
- Figure 6 shows how actions at a given time can be traced back to coin-toss roots or origin roots.
- Coin-toss roots are the source of experimentation; imitation spreads their consequences through the network.

Economic message: Random experimentation is not just noise. It is the seed that makes universal diffusion bounds possible.

6 Short summary

- The paper studies diffusion of a superior innovation in social networks with coordination payoffs and random payoff shocks.
- Agents update asynchronously according to Poisson clocks and choose perturbed best responses.
- The adoption probability is summarized by $r(x) = 1 - F(c - a - 2cx)$.
- The supporting line $L(x)$ below $r(x)$ is the central analytical object.
- If $L(0) > 0$ and the target q is below the fixed point p of L, diffusion is fast in expectation.
- The main waiting-time bound is independent of network topology, size, degree distribution, directionality, and time variation.
- The proof uses a lower-bound linear process that can be interpreted as coin-toss experimentation plus imitation.
- An imitation forest tracks the sources of agents' actions.
- Feige's inequality supplies a topology-free lower bound for weighted sums of Bernoulli coin tosses.
- The analysis extends to multiple competing technologies using a worst-case response function $\rho(x)$.
- For log-concave shocks, the worst competitor to the best technology is the second-best technology.
- Regular networks have sharper convergence rates and strong persistence.
- Star networks can converge quickly but have weak persistence because the hub matters so much.
- If $r(q) < q$, convergence can be exponentially slow in regular networks with growing degree.

7 Conclusion

7.1 Core conclusion

Arieli, Babichenko, Peretz, and Young show that innovation diffusion can be guaranteed to reach a target adoption share in bounded expected time even when the network is arbitrary, directed, irregular, and evolving. The universal bound depends on the response function, not on the network.

7.2 Economic intuition

The intuition is that random experimentation creates initial adopters, and imitation spreads their actions. The supporting line $L(x)$ gives a conservative linear approximation to the true response function. If this linear approximation has positive intercept, then even the slow lower-bound process eventually generates enough successful coin-toss roots to cross the target threshold.

7.3 Takeaway

The paper separates what must be known from what need not be known. One does not need the full social network to bound diffusion speed. One needs enough behavioral information to verify that the response function has a positive lower linear support and that the desired target lies below its fixed point.